

LOKESH R M

AI / ML Engineer · Agentic AI, Computer Vision, Edge AI, Python

+91 9035446966 | hi.lokeshrm@gmail.com | lokeshrm.work@gmail.com | Bengaluru, Karnataka, India
linkedin.com/in/thisislokeshrm | github.com/hilokeshrm | hilokeshrm.github.io

SUMMARY

AI Engineer at Sirena Technologies building production agentic AI and robotics systems. Architected Agent OS, a 53 service multi agent platform running at 3 enterprise clients, and own the Policy Gateway enforcing 153 policies across it. Work spans LLMs, computer vision, voice pipelines and edge inference, with cloud services supporting 80+ concurrent robots.

EXPERIENCE

AI Engineer - Sirena Technologies, Bengaluru, India

Jul 2026 - Present

- Architected **Agent OS**, a 53 service multi agent platform deployed to 3 enterprise clients, letting in house and third party agents interoperate to model business functions for small and medium sized businesses. Platform team of 12.
- Designed and own the **Policy Gateway**, enforcing 153 policies for centralised agent authorisation, control and monitoring. Led a team of 3.
- Led development of 5 in house agents across opportunity tracking, ROI, marketing and finance workflows, and 3 third party agent integrations.
- Led a 5 person team building **Sirena Cloud**, the voice, vision and LLM service layer supporting 80+ concurrent robots on an NVIDIA RTX Pro 6000.
- Built a streaming voice pipeline combining VAD, ASR, TTS and LLM reasoning at roughly 60 ms response latency.
- Fine tuned YOLO and SSD detectors for office and industrial environments, and optimise inference with ONNX for NVIDIA Jetson Orin Nano.
- Self host open weight LLMs, build RAG retrieval on pgvector, and deploy ROS 2 Nav2 and SLAM navigation for autonomous service robots.

Robotics Engineer, Graduate Trainee - Sirena Technologies, Bengaluru and Solan, India

May 2025 - Jun 2026

- Built MATLAB and Python locomotion simulations for biped and quadruped robots covering walk sequences and control algorithms, and validated every platform in simulation before hardware build.
- Developed features in the LS6 customised robot controller codebase and implemented ROS 2 motion control for robotic arms and wheeled robots.
- Led a 3 person team building the cloud system for an ESP32 P4 based home robotics product, including ESP-IDF firmware integration.
- Completed the graduate engineering rotation at the Solan facility and converted to a full time engineering role.

Freelance Full Stack and AI Engineer - Independent, Bengaluru, India

2024 - Present

- Delivered production web and AI applications for 5+ clients, covering LLM integration, agent workflows, APIs and backend infrastructure.

Machine Learning Intern - MURTI Labs, GITAM Bengaluru, India

May 2024 - Apr 2025

- Trained LSTM based object detection and action recognition models on drone video to 83% accuracy, and implemented path mapping for autonomous delivery and emergency response drones.
- Built the image processing and data collection pipeline feeding model training.

SKILLS

AI and Machine Learning: LLMs, RAG, Multi Agent Systems, Agent Orchestration, PyTorch, TensorFlow, Scikit-learn, LSTM, pgvector, Open Weight Model Hosting

Agent Frameworks: LangGraph, LangChain, CrewAI, AutoGen, MCP

Computer Vision and Speech: YOLO, SSD, OpenCV, MediaPipe, ONNX, ASR, TTS, Voice Activity Detection

Robotics and Edge AI: ROS 2, Nav2, SLAM, NVIDIA Jetson Orin Nano, RTX Pro 6000, NVIDIA Isaac Sim, ESP32 P4, Edge Inference

Infrastructure: Docker, Kubernetes, FastAPI, Kafka, Redis, Google Cloud, PostgreSQL, MongoDB, Linux, Git

Programming: Python, C++, C, MATLAB, SQL, JavaScript

PROJECTS

Secure Federated Learning with Blockchain

Sep 2024

Federated training system letting organisations train shared models on decentralised data, with blockchain backed coordination and data provenance so raw records never leave the owner.

Stack: Python, TensorFlow, Hyperledger Fabric, Solidity, Homomorphic Encryption, Differential Privacy

EDUCATION

B.Tech, Computer Science and Engineering - GITAM (Deemed to be University), Bengaluru

Aug 2022 - Aug 2026

CGPA: 7.9 / 10

ACHIEVEMENTS

- INSPIRE Award MANAK** (Government of India, state level) · **Young Scientist Award** (Government of Karnataka, district level) · **Rajya Puraskar** (Governor's Award) · Finalist, Internal Smart India Hackathon, GITAM
- Secretary, IEEE Computer Society GITAM** (Dec 2024 - Aug 2025) · Student Placement Coordinator, GITAM Career Guidance Cell